

Lab 02 - Expected evidence

UIDs, ages, IPs and node names will differ. The strings and structure below are what a correct run looks like.

1. Authoring aids

```
$ kubectl run depot-web --image=nginx:1.27-alpine --port=80 --dry-run=client -o yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: depot-web
  name: depot-web
spec:
  containers:
  - image: nginx:1.27-alpine
    name: depot-web
    ports:
    - containerPort: 80
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

Note resources: {} - the generator never writes requests or limits. That is why you finish the manifest by hand.

2. ConfigMap built from data/

```
$ kubectl -n kcna-lab02 create configmap depot-content \
  --from-file=index.html=data/depot-status.html \
  --from-file=depots.csv=data/depots.csv
configmap/depot-content created

$ kubectl -n kcna-lab02 get configmap depot-content -o jsonpath='{range $.data}{"\n"}{end}' >/dev/null
$ kubectl -n kcna-lab02 describe configmap depot-content | head -n 8
Name:          depot-content
Namespace:     kcna-lab02
Labels:        <none>
Annotations:   <none>

Data
====
depots.csv:
```

3. The Pod

```
$ kubectl apply -f manifests/20-depot-web-pod.yaml
pod/depot-web created

$ kubectl -n kcna-lab02 get pod depot-web -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP           NODE
NOMINATED NODE READINESS GATES
depot-web     1/1     Running   0           18s   10.244.0.11  kcna-qa-20260905-control-plane
<none>        <none>
```

READY 1/1 means the readinessProbe against /index.html succeeded.

4. Server-side defaulting

```
$ kubectl -n kcna-lab02 get pod depot-web -o jsonpath='{.spec.dnsPolicy}{"\t"}{.spec.serviceAccountName}{"\t"}{.spec.schedulerName}{"\n"}'
ClusterFirst default default-scheduler
```

None of those three fields were in your manifest.

5. Content actually served

```
$ kubectl -n kcna-lab02 exec depot-web -c web -- wget -qO- http://127.0.0.1/index.html | head -n 3
<!DOCTYPE html>
<html lang="en">
<head>

$ kubectl -n kcna-lab02 exec depot-web -c web -- wget -qO- http://127.0.0.1/depots.csv | head -n 3
depot_code,depot_name,region,bays,cold_chain,shift_pattern,last_audit
MF-SIN-01,Jurong Port Depot,Singapore West,24,yes,3x8,2026-07-14
MF-SIN-02,Tuas Megahub,Singapore West,48,yes,3x8,2026-08-02
```

Via port-forward from your workstation:

```
$ kubectl -n kcna-lab02 port-forward pod/depot-web 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
```

```
$ curl -s http://127.0.0.1:8080/ | grep STATUS
<p class="ok">STATUS: OK - served from a Pod in namespace kcna-lab02</p>
```

6. Failure injection - ImagePullBackOff

```
$ kubectl apply -f manifests/90-depot-web-badtag.yaml
pod/depot-web-badtag created

$ kubectl -n kcna-lab02 get pod depot-web-badtag
NAME                READY   STATUS    RESTARTS   AGE
depot-web-badtag    0/1     ErrImagePull    0           8s

$ kubectl -n kcna-lab02 get pod depot-web-badtag
NAME                READY   STATUS             RESTARTS   AGE
depot-web-badtag    0/1     ImagePullBackOff    0          45s
```

```
$ kubectl -n kcna-lab02 describe pod depot-web-badtag | tail -n 9
```

Events:	Type	Reason	Age	From	Message
	Normal	Scheduled	62s	default-scheduler	Successfully assigned kcna-lab02/depot-web-badtag to kcna-qa-20260905-control-plane
	Normal	Pulling	22s (x3 over 62s)	kubelet	Pulling image "nginx:1.27-alpin"
	Warning	Failed	21s (x3 over 61s)	kubelet	Failed to pull image "nginx:1.27-alpin": failed to pull and unpack image "docker.io/library/nginx:1.27-alpin": failed to resolve reference "docker.io/library/nginx:1.27-alpin": docker.io/library/nginx:1.27-alpin: not found
	Warning	Failed	21s (x3 over 61s)	kubelet	Error: ErrImagePull
	Normal	BackOff	8s (x4 over 60s)	kubelet	Back-off pulling image "nginx:1.27-alpin"
	Warning	Failed	8s (x4 over 60s)	kubelet	Error: ImagePullBackOff

```
$ kubectl -n kcna-lab02 get pod depot-web-badtag \
-o jsonpath='{.status.containerStatuses[0].state.waiting.reason}'
ImagePullBackOff
```

7. Verification script

```
$ bash verification/checks.sh

== 0. Cluster reachability
[PASS] kubectl can reach the API server

== 1. Namespace
[PASS] namespace kcna-lab02 exists

== 2. ConfigMap depot-content is built from data/
[PASS] ConfigMap depot-content exists (keys: depots.csv index.html )
[PASS] key index.html present
[PASS] key depots.csv present
[PASS] ConfigMap depots.csv row count matches data/depots.csv (10 depots)

== 3. Pod depot-web
[PASS] Pod depot-web phase is Running
[PASS] container 'web' passed its readinessProbe (ready=true)
[PASS] image is pinned to nginx:1.27-alpine (no :latest)
[PASS] containerPort 80 declared and named 'http'
[PASS] explicit resources set (requests.cpu=50m, limits.memory=128Mi)

== 4. Content is served from the mounted ConfigMap
[PASS] GET /index.html returns the Meridian Freight page
[PASS] GET /depots.csv returns the synthetic depot dataset

== 5. Server-side defaulting is observable
[PASS] API server defaulted dnsPolicy=ClusterFirst and serviceAccountName=default

== 6. Failure-injection Pod (only if you left it in place)
[PASS] depot-web-badtag is in ImagePullBackOff as designed

Result: 14 passed, 0 failed
```

If you already deleted depot-web-badtag, section 6 prints [SKIP] depot-web-badtag not present and the total is 13 passed, 0 failed.